

AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

Sessional Test

Program: B.Tech
 Session: 2025-26
 Subject: Energy Science & Engineering
 Max. Marks: 50

Semester: III
 Section: IT, CSIT, ME, EN, EC, CE
 Subject Code: BOE-404
 Time: 2 Hours

OBE Remarks:

Q.No	1	2	3	4	5	6	7	8	9	10	11	12
CO No.	CO1	CO1	CO2	CO2	CO2	CO1	CO1	CO1	CO2	CO2	CO1	CO2
Bloom's Level* (L1 to L6)	L2	L2	L3	L3	L1	L4	L2	L2	L3	L2	L2	L2
Weightage CO1: 26.5						Weightage CO2: 23.5						

*Bloom's Level: L1: Remember, L2: Understand, L3: Apply, L4: Analyze, L5: Evaluate, L6: Create

Note: Answer all the sections with all the questions.

Section-A

(2*5=10)

1. Describe photo and thermal phenomena in solid state.
2. Explain the difference between gas power cycle and steam power cycle.
3. Illustrate the difference between nuclear fission and nuclear fusion process.
4. Enlist the advantages of nuclear energy.
5. Define binding energy & mass defect.

Section-B

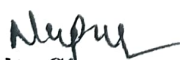
(5*5=25)

6. Analyze and contrast the working principles for refrigeration systems and heat pumps, discussing the thermodynamics process behind each.
7. Explain the principle of Steam and Gas power cycles with neat sketch.
8. Explain the conversion of heat energy to mechanical energy with an example.
9. Draw the binding energy curve showing variation of binding energy per nucleon with mass number. With the help of this explain the phenomenon of nuclear fusion and fission & stability concept of nuclei.
10. Explain the four fundamental forces of nature and discuss their relative strengths and ranges.

Section-C

(7.5*2=15)

11. Describe the construction, working, and efficiency of a Carnot heat engine with diagrams.
12. Explain in detail the layout, working and operation of a nuclear fission reactor.


 Faculty Sign


 HoD Sign